

2.1. Design and Performance Evaluation of a Chebyshev Type-I IIR Low-Pass Filter using MATLAB

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clc; clear;

Wp=0.4; Ws=0.55; Rp=1; As=40;

[N,Wn]=cheb1ord(Wp,Ws,Rp,As);

[b,a]=cheby1(N,Rp,Wn);

fprintf('Filter Order = %d\n',N);

fprintf('Cutoff Frequency = %.2f * pi rad/sample\n',Wn);

if all(abs(roots(a))<1)

disp('System is STABLE');

else

disp('System is UNSTABLE');

end

% ----- GRAPHS -----

figure; freqz(b,a); % Magnitude + Phase

figure; impz(b,a); % Impulse Response

figure; grpdelay(b,a); % Group Delay

figure; zplane(b,a); % Pole-Zero
```





